

Since $P = I^2 R = V^2 / R$, the power-time graph will be a cosine-square curve.

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Ans: B

Using $P_0 = \frac{V_0^2}{R}$,

Since the transformer is ideal, $P_{\text{primary}} = P_{\text{secondary}}$

For primary coil, $\frac{P_{o,initial}}{P_{o,final}} = \left(\frac{V_{0,initial}}{V_{0,final}} \right)^2 \rightarrow \frac{36 \times 2}{P_{o,final}} = \left(\frac{2.8}{1.4} \right)^2$
 $P_{o,final} = 18 \text{ W}$

$$I_0 = \frac{P_0}{V_0} = \frac{18}{1.4} = 12.86 \text{ A}$$

$$I_{rms} = \frac{12.86}{\sqrt{2}} = 9.09 \text{ A}$$

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Ans: A